

IMAT MOCK EXAM 10 (Legacy): 2026 Blueprint

60 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · personal target: 44/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. Screening programmes aim to detect disease in people who have no symptoms, on the assumption that treating earlier improves outcomes. Yet early detection has a structural drawback. Some of the cancers that screening finds grow so slowly that they would never have caused symptoms in a person's lifetime; counting these cases as cures makes screening appear more effective than it really is, a phenomenon known as overdiagnosis. At the same time, trials show that breast cancer screening does reduce deaths from breast cancer, though the absolute benefit for any individual woman is small, because the disease is uncommon in any single year. Deciding whether to attend screening therefore involves weighing a small reduction in the risk of death against the possibility of unnecessary treatment.

Which of the following statements is most strongly supported by the passage?

- A)** Screening programmes can only succeed against diseases that produce symptoms at an early stage.
- B)** Overdiagnosis occurs because screening detects some cancers that would never have harmed the person.
- C)** The absolute benefit of breast cancer screening is large, because the disease is common.
- D)** Breast cancer screening has been shown to reduce deaths from all causes.
- E)** The passage implies that women should generally be advised not to attend screening.

Answer: B

The passage defines overdiagnosis in exactly these terms: some cancers found by screening grow so slowly that they would never have caused symptoms or harmed the person in their lifetime, so that statement is directly supported. The text limits the proven benefit to deaths from breast cancer, not deaths from all causes, and it calls the absolute benefit small because the disease is uncommon in any single year, so the claim of a large benefit from a common disease reverses the passage. Screening targets people without symptoms, and the passage presents a weighing of small benefit against possible harm rather than any advice to avoid screening.

2. Professional ecologists cannot survey species across entire countries with the staff and funding available to them. Citizen science projects address this gap by recruiting volunteers to record what they observe: birds counted in gardens, butterflies photographed on walks, plants logged along roadsides. The resulting datasets have repeatedly revealed changes in the distribution and timing of species, such as earlier flowering, that smaller professional studies would have missed. The approach has limits, however. Volunteers sample unevenly, recording more near cities and at weekends, and they more easily notice conspicuous species. Project organisers therefore write structured protocols, provide identification training and statistically adjust for sampling bias, treating volunteer data not as a cheap substitute for professional monitoring but as a complement that extends its geographical reach.

Which of the following statements follows from the passage?

- A)** Citizen science projects can extend the geographical coverage of monitoring beyond what professional ecologists could achieve alone.
- B)** Data collected by volunteers are too unreliable to inform ecological research.
- C)** Volunteers overlook conspicuous species because these are difficult to photograph.
- D)** Statistical adjustment eliminates sampling bias from volunteer-collected data.
- E)** Professional ecologists have largely stopped carrying out their own surveys since volunteers took over the work.

Answer: A

The passage states that professional ecologists cannot survey entire countries and that organisers treat volunteer data as a complement that extends the geographical reach of monitoring, which is precisely the claim that citizen science extends coverage beyond what professionals could achieve alone. The claim that volunteer data are too unreliable is contradicted by the datasets repeatedly revealing real changes such as earlier flowering, and the text calls volunteers a complement, not a substitute that replaced professional surveys. The idea that volunteers overlook conspicuous species inverts the statement that they notice such species more easily, and organisers statistically adjust for sampling bias, which is not the same as eliminating it.

3. (General knowledge) The chorus *Va, pensiero*, known in English as the Chorus of the Hebrew Slaves and often treated as an unofficial Italian anthem, comes from the opera *Nabucco*, first performed at La Scala in 1842. Who was the composer?

- A)** Gioachino Rossini
- B)** Giacomo Puccini
- C)** Vincenzo Bellini
- D)** Giuseppe Verdi
- E)** Gaetano Donizetti

Answer: D

Nabucco, premiered at La Scala in Milan in 1842, was composed by Giuseppe Verdi, and its third-act chorus *Va, pensiero*, the Chorus of the Hebrew Slaves, became an unofficial Italian anthem and established Verdi's reputation. Bellini, Donizetti and Rossini belong to earlier generations of Italian opera and Puccini to the following generation, so none of them matches the 1842 premiere.

4. When Alfred Wegener proposed continental drift in 1912, most geologists dismissed him. He had assembled an impressive case: the fit of the Atlantic coastlines, identical fossils on opposite shores, and traces of ancient glaciation in what are now tropical latitudes. What he lacked was a credible driving force, and his suggestion that continents plough through the ocean floor was physically implausible. The balance changed only in the decades after the Second World War, when mapping of the ocean floor revealed mid-ocean ridges and magnetic stripes recording seafloor spreading, and when the locations of earthquakes traced out the boundaries of moving plates. Drifting continents, once rejected, became the organising principle of modern geology not because Wegener's evidence was rediscovered but because new kinds of evidence, gathered with instruments he never had, settled the question of mechanism.

Which of the following statements is supported by the passage?

- A) Ocean-floor mapping showed that mid-ocean ridges and magnetic stripes do not exist.
- B) The question of what drives continental movement remains unanswered in modern geology.
- C) Plate tectonics won acceptance once Wegener's original evidence was rediscovered after the Second World War.
- D) Wegener's theory was rejected mainly because the observations behind it were later shown to be mistaken.
- E) The mechanism Wegener proposed for continental movement was regarded as physically implausible.

Answer: E

The passage states directly that Wegener's own suggestion, that continents plough through the ocean floor, was physically implausible, which is exactly the claim about his proposed mechanism. His observational evidence is called an impressive case, so rejection was not because his observations were later shown wrong, and acceptance came after the Second World War from new kinds of evidence (mid-ocean ridges, magnetic stripes, earthquake locations), explicitly not from rediscovering his data. The new evidence settled the question of mechanism, so that question is not still open.

5. All the guides at the city museum speak at least three languages. Nobody who speaks at least three languages ever needs an interpreter on a tour. Which one of the following statements must be true?

- A) Everyone who speaks at least three languages works as a guide at the city museum.
- B) Every guide at the city museum works full time.
- C) No guide at the city museum ever needs an interpreter on a tour.
- D) Some guides at the city museum need an interpreter on some tours.
- E) Anyone who never needs an interpreter on a tour speaks at least three languages.

Answer: C

Chain the premises: every guide speaks at least three languages, and nobody who speaks at least three languages ever needs an interpreter on a tour, so no guide at the museum ever needs an interpreter on a tour. The reversals of the premises, that every trilingual speaker is a museum guide or that anyone spared an interpreter must speak three languages, are illegitimate converses and do not follow, and the claim that some guides need an interpreter contradicts the chain outright. Working hours are never mentioned in the premises.

6. Five colleagues, Elena, Jonas, Nadia, Omar and Priya, sit around a circular table with five chairs numbered 1 to 5 in clockwise order, one person per chair. Elena sits in chair 1. Nadia sits immediately clockwise from Elena. Omar does not sit next to Elena. Jonas sits immediately anticlockwise from Priya. Which one of the following statements must be true?

- A) Omar sits next to Elena.
- B) Nadia and Priya sit next to each other.
- C) Omar sits between Nadia and Jonas.
- D) Priya sits immediately clockwise from Elena.
- E) Jonas sits immediately anticlockwise from Elena.

Answer: C

Step 1: Elena is in chair 1 and Nadia immediately clockwise from her, so Nadia is in chair 2, leaving chairs 3, 4 and 5 for Jonas, Omar and Priya. Step 2: Jonas immediately anticlockwise from Priya allows only Priya 5 with Jonas 4, or Priya 4 with Jonas 3; the second case forces Omar into chair 5, which is adjacent to Elena's chair 1, violating the condition on Omar. Step 3: the seating is uniquely Elena 1, Nadia 2, Omar 3, Jonas 4, Priya 5. Omar in chair 3 sits between Nadia and Jonas; Priya sits immediately anticlockwise, not clockwise, from Elena; Omar is not next to Elena; Nadia and Priya are not neighbours; and the chair immediately anticlockwise from Elena holds Priya, not Jonas.

7. Five coloured balls, one each of blue, green, red, white and yellow, are placed in five boxes numbered 1 to 5 from left to right, one ball per box. The green ball is immediately to the left of the red ball. The red ball is in a box with an even number. The yellow ball is somewhere to the right of the green ball. The white ball is immediately to the right of the yellow ball. The blue ball is in box 5. In which box must the yellow ball be?

- A) Box 3 B) Box 1 C) Box 5 D) Box 4 E) Box 2

Answer: A

Step 1: red is in an even-numbered box with green immediately to its left, so the pair occupies boxes 1 and 2, or boxes 3 and 4. Step 2: if green were in box 3, yellow would need a box to the right of it, but boxes 4 and 5 are taken by red and blue while boxes 1 and 2 lie to the left, so the pair must be green 1, red 2. Step 3: yellow and white then fill boxes 3 and 4, and white being immediately to the right of yellow places yellow in box 3 and white in box 4.

8. One term is missing from the following sequence: 5, 8, 14, ?, 35. What is the missing term?

- A) 18 B) 20 C) 22 D) 23 E) 26

Answer: D

Step 1: the successive differences are $8 - 5 = 3$ and $14 - 8 = 6$, and the pattern of differences growing by 3 each time gives 3, 6, 9, 12. Step 2: the missing term is $14 + 9 = 23$, and the check $23 + 12 = 35$ matches the given final term exactly. Doubling the differences instead (3, 6, 12, 24) would make the fifth term 50 rather than 35, so the constant growth of 3 is the only pattern that fits.

9. What is the angle between the hour hand and the minute hand of an analogue clock at 5:30?

- A) 5 degrees B) 15 degrees C) 25 degrees D) 30 degrees E) 45 degrees

Answer: B

At 5:30 the minute hand points exactly at the 6, that is 180 degrees from the 12. The hour hand moves 0.5 degrees per minute, so half an hour past 5 it sits at $150 + 15 = 165$ degrees.

The angle between the hands is $180 - 165 = 15$ degrees

The value of 30 degrees would hold only exactly on the hour, forgetting that the hour hand has moved halfway toward the 6.

Section B - Biology (Q10-32)

10. Glycolysis is often divided into an energy investment stage and an energy payoff stage. Which of the following statements about the breakdown of one molecule of glucose to pyruvate is true?

- A)** ATP is invested to pump pyruvate across the mitochondrial membrane before the payoff stage can begin.
- B)** Four ATP molecules are consumed in the investment stage and only two are produced in the payoff stage, so glycolysis causes a net loss of ATP.
- C)** The investment stage can proceed only when oxygen is available to withdraw electrons from the pathway.
- D)** The net yield is four ATP per glucose, because each ATP forming reaction occurs only once per glucose molecule.
- E)** The two ATP molecules are spent before the six carbon sugar is cleaved into two three carbon molecules, so every reaction of the payoff stage happens twice for each glucose that enters.

Answer: E

In the investment stage, hexokinase and phosphofructokinase each spend one ATP while the sugar still has all six carbons; aldolase then cleaves it into two three-carbon sugars. Every payoff-stage reaction, including both ATP-producing steps, therefore runs twice per glucose, so four ATP are produced against two invested, a net gain of two. Claims of a net ATP loss or of a four-ATP net yield both mangle this accounting, and glycolysis needs no oxygen at any step, nor does it pump pyruvate across any membrane.

11. In eukaryotic cells, the pyruvate produced by glycolysis does not enter the Krebs cycle directly. Which statement describes what happens to it first?

- A)** It is converted to lactate inside the mitochondrial matrix so that NAD^+ can be regenerated for glycolysis.
- B)** It crosses both mitochondrial membranes unchanged and condenses directly with oxaloacetate to form citrate.
- C)** In the mitochondrial matrix it is oxidatively decarboxylated: one carbon leaves as CO_2 , NAD^+ is reduced to NADH , and the remaining two carbon acetyl group is attached to coenzyme A.
- D)** It is converted to acetyl-CoA in the cytosol, and the acetyl-CoA is then carried into the mitochondria by transport vesicles.
- E)** It is fully oxidised to CO_2 and water at the inner mitochondrial membrane by the cytochromes of the electron transport chain.

Answer: C

Pyruvate crosses into the mitochondrial matrix, where the pyruvate dehydrogenase complex oxidatively decarboxylates it: one carbon leaves as CO_2 , NAD^+ is reduced to NADH , and the remaining two-carbon acetyl group is transferred to coenzyme A, forming acetyl-CoA. Only acetyl-CoA, never pyruvate itself, condenses with oxaloacetate to make citrate. Lactate formation is a cytosolic fermentation route, secretory traffic does not move metabolites between compartments by vesicles, and the electron transport chain oxidises NADH and FADH_2 , not pyruvate.

12. Which of the following statements about the Krebs (citric acid) cycle is true?

- A)** Oxaloacetate is consumed in stoichiometric amounts, so the diet must continuously supply it for the cycle to keep turning.
- B)** For each acetyl group oxidised, one turn of the cycle releases four molecules of CO₂.
- C)** Most of the ATP generated in aerobic respiration is produced by substrate level phosphorylation inside the cycle itself.
- D)** One reaction of the cycle consumes molecular oxygen directly, which is what links the cycle to the electron transport chain.
- E)** Each turn regenerates the four carbon acceptor oxaloacetate, so the cycle can continue as long as acetyl-CoA, NAD⁺, FAD and ADP are supplied.

Answer: E

Each turn of the cycle regenerates the four-carbon acceptor oxaloacetate, so the cycle runs catalytically and needs only a supply of acetyl-CoA plus oxidised cofactors (NAD⁺, FAD) and ADP. Two CO₂, not four, are released per acetyl group, no cycle reaction consumes molecular oxygen (that happens in the electron transport chain), and the single GTP made by substrate level phosphorylation per turn is a minor share of the ATP of aerobic respiration, most of which comes from oxidative phosphorylation. Oxaloacetate is regenerated each turn and does not need to be eaten.

13. In the mitochondrial electron transport chain, which complexes pump protons (H⁺) from the matrix to the intermembrane space?

- A)** Complexes II and III are the only pumping complexes, because electrons reach both of them at the same energy level.
- B)** Complexes I, III and IV pump protons, while complex II passes electrons to ubiquinone without pumping any.
- C)** Only complex IV pumps protons, and it does so while reducing oxygen to water.
- D)** All four complexes pump protons, which is why NADH and FADH₂ yield exactly the same number of ATP.
- E)** No complex pumps protons directly; protons cross the membrane together with the electrons inside the mobile carriers.

Answer: B

Complexes I, III and IV use the energy of electron transfer to pump H⁺ from the matrix to the intermembrane space, whereas complex II, succinate dehydrogenase, merely passes electrons from FADH₂ to ubiquinone without pumping. This is exactly why FADH₂, which enters at complex II and bypasses complex I, yields less ATP than NADH; if all four complexes pumped, the two donors would give identical yields, which they do not. Protons never ride across the membrane together with the electrons inside the mobile carriers.

14. Actively respiring mitochondria maintain a pH difference between the matrix and the intermembrane space. Which statement is correct?

- A)** The intermembrane space is more acidic than the matrix, because electron transport pumps H^+ out of the matrix and the resulting gradient drives H^+ back through ATP synthase.
- B)** The intermembrane space is more alkaline, because ATP synthesis consumes protons inside the matrix faster than electron transport can replace them.
- C)** The two compartments have identical pH values, because protons diffuse freely through the outer mitochondrial membrane until equilibrium is reached.
- D)** No significant pH difference can build up, because the inner mitochondrial membrane is freely permeable to protons.
- E)** The matrix is more acidic than the intermembrane space, because ATP synthase releases H^+ into the matrix as it converts ADP and phosphate into ATP.

Answer: A

Electron transport pumps H^+ out of the matrix into the intermembrane space, so the intermembrane space becomes more acidic (lower pH) than the matrix, and the stored electrochemical gradient drives H^+ back inward through ATP synthase to power ATP synthesis. The mechanism depends on the inner membrane being impermeable to protons; free proton permeability would collapse the gradient, and a more alkaline intermembrane space would reverse the actual direction of pumping.

15. During a 400 m sprint, fast muscle fibres convert pyruvate to lactate. Which statement correctly describes the subsequent fate of this lactate?

- A)** It is converted back into glucose inside the same muscle fibre and immediately exported into the blood.
- B)** It remains locked inside the fibre as a permanent inert waste product.
- C)** It is excreted unchanged by the kidneys, which greatly lowers the pH of the urine after every sprint.
- D)** Much of it is carried in the blood to the liver, where it is converted back into glucose that can be returned to the blood or stored as liver glycogen.
- E)** It is converted by human muscle enzymes into ethanol and CO_2 , exactly as in yeast fermentation.

Answer: D

In the Cori cycle the blood carries lactate from the muscle to the liver, which converts it back into glucose by gluconeogenesis and returns the glucose to the blood or stores it as liver glycogen. Muscle fibres cannot do this themselves because they lack glucose 6 phosphatase, so they cannot export free glucose made from lactate, and human cells never ferment lactate to ethanol and CO_2 as yeast does. Lactate is a temporary carbon carrier, not an inert waste locked in the fibre, and it is not simply excreted by the kidneys.

16. The rate of an enzyme catalysed reaction is measured with and without inhibitor X at a series of substrate concentrations. X lowers the rate at every concentration, but at very high substrate concentration the inhibited reaction approaches the same maximum velocity as the uninhibited reaction. Which conclusion is correct?

- A)** X is a competitive inhibitor: the apparent K_m is raised but V_{max} is unchanged, because sufficiently concentrated substrate outcompetes X for the active site.
- B)** X binds only to the enzyme substrate complex, so its inhibitory effect becomes stronger as substrate concentration rises.
- C)** X works by raising the activation energy of the catalysed reaction, so no concentration of substrate can overcome it.
- D)** X irreversibly destroys the active sites, so adding more substrate can never bring the rate back toward the uninhibited maximum.
- E)** X is a non-competitive inhibitor that lowers V_{max} , and the apparent convergence of the two curves must be an experimental artefact.

Answer: A

The rate falling at every substrate concentration yet approaching the same maximum velocity at very high concentration is the signature of competitive inhibition: substrate and inhibitor compete for the same active site, and sufficiently concentrated substrate outcompetes the inhibitor, leaving V_{max} unchanged while the apparent K_m rises. A non-competitive inhibitor removes functional enzyme and lowers V_{max} in a way no substrate concentration can overcome, so the observed convergence at the uninhibited maximum rules that out, as do irreversible destruction of active sites and inhibition that grows with substrate concentration.

17. The vitamin niacin (vitamin B3) is a precursor of the nicotinamide ring of NAD^+ . Severe dietary deficiency of niacin would be expected to impair most directly the activity of:

- A)** hydrolytic enzymes such as amylase and trypsin, because NAD^+ supplies the water used to break glycosidic and peptide bonds.
- B)** dehydrogenases of pathways such as glycolysis and the Krebs cycle, because NAD^+ is a coenzyme that accepts electrons from oxidised substrates and must be continuously regenerated.
- C)** only the protein complexes of the electron transport chain, because NAD^+ is split there to release the energy stored in ATP.
- D)** the sodium potassium pump, because NAD^+ is hydrolysed to provide the energy for active transport.
- E)** every enzyme in the body, because NAD^+ is needed for the correct folding of all newly made proteins.

Answer: B

Niacin is the precursor of the nicotinamide ring of NAD^+ , the reusable coenzyme that dehydrogenases use to accept electrons from substrates being oxidised in glycolysis, the link reaction and the Krebs cycle, so niacin deficiency cripples those dehydrogenases most directly. ATP, not NAD^+ , powers the sodium potassium pump; water for hydrolysis comes from the solvent, not from NAD^+ ; NAD^+ has no general role in protein folding; and the electron transport chain regenerates NAD^+ rather than splitting it for energy.

18. According to the sliding filament model, which of the following occurs within a sarcomere during a strong contraction of a skeletal muscle fibre?

- A)** The thick filaments shorten because the myosin tails coil up like springs around the actin filaments.
- B)** The thin filaments shorten because actin monomers are stripped from their ends during force generation.
- C)** The H zone widens as the thin filaments are pulled away from the centre of the sarcomere.
- D)** The thick and thin filaments keep their original lengths while the sarcomere shortens: the I band and the H zone become narrower but the A band keeps its width.
- E)** The A band and the I band shorten in the same proportion, reflecting the equal shortening of thick and thin filaments.

Answer: D

In the sliding filament model, cross-bridge cycling slides the thick and thin filaments past each other without changing either filament's length, so the sarcomere shortens while the A band, whose width equals the thick filament length, keeps its width. The I band and the H zone narrow as filament overlap increases. Neither filament coils like a spring or loses monomers during contraction, and pulling thin filaments away from the centre would widen, not narrow, the H zone.

19. Which of the following statements about motor units in skeletal muscle is true?

- A)** Force can be increased only by raising the firing frequency of a fixed set of motor units, never by recruiting additional units.
- B)** During a weak contraction, all of the muscle's motor units are active, but each one fires action potentials of reduced amplitude.
- C)** A motor unit consists of one motor neuron together with all the muscle fibres it innervates, and when that neuron fires, every fibre in the unit contracts.
- D)** Muscles used for fine, precise movements, such as the external muscles of the eye, are organised into a small number of large motor units.
- E)** A motor unit consists of a single muscle fibre together with all the motor neurons that supply it.

Answer: C

A motor unit is one motor neuron together with all the muscle fibres it innervates, and because a neuron's signal is all-or-none, every fibre of the unit contracts when that neuron fires. Fine, precise control, as in the external eye muscles, requires many small motor units rather than a few large ones; whole-muscle force is graded both by firing frequency and by recruiting additional units; and individual action potentials do not shrink in amplitude during weak contractions.

20. Which of the following correctly matches an epithelial type with its normal location and the advantage it provides there?

- A)** Ciliated pseudostratified columnar epithelium; the walls of the alveoli; the beating cilia stir the air to speed diffusion.
- B)** Simple squamous epithelium; the lining of the urinary bladder; its many cell layers protect against acidic urine.
- C)** Transitional epithelium; the lining of the small intestine; its ability to stretch provides a huge absorptive surface.
- D)** Simple columnar epithelium; the dry outer surface of the skin; its single layer heals wounds faster than any multilayered epithelium.
- E)** Keratinised stratified squamous epithelium; the epidermis of the skin; its dead, keratin filled surface cells resist abrasion and reduce water loss.

Answer: E

The epidermis is keratinised stratified squamous epithelium: its dead, keratin-filled surface cells resist abrasion and reduce water loss from the dry outer skin, which is exactly the stated match. Transitional (stretchable) epithelium lines the urinary bladder, not the small intestine; alveolar walls are lined by a thin simple squamous layer with no cilia; and skin's protection comes from having many layers, not from healing faster as a single layer.

21. Which of the following statements about connective tissues is true?

- A)** Adipose tissue stores fat droplets in a sparse network of collagen fibres rather than inside cells.
- B)** Cartilage contains no blood vessels: its chondrocytes sit in matrix filled spaces called lacunae and receive nutrients by diffusion, which is one reason injured cartilage heals slowly.
- C)** The collagen fibres of tendons are secreted by chondrocytes that migrate in from nearby cartilage.
- D)** Dense regular connective tissue such as tendon is rich in ground substance and poor in fibres, which makes it highly elastic.
- E)** Blood is not classified as connective tissue because it contains no extracellular matrix.

Answer: B

Cartilage is avascular: its chondrocytes sit in matrix-filled lacunae and receive nutrients by diffusion, which is one reason injured cartilage heals slowly. Blood is a connective tissue whose matrix is plasma, tendon collagen is secreted by fibroblasts (not by cartilage cells) and dense regular tissue is fibre-rich and stiff rather than ground-substance rich and elastic, while adipose tissue stores fat inside adipocytes, not in an extracellular fibre network.

22. A protein is about to be secreted from a pancreatic cell. Which statement correctly describes its passage through the Golgi apparatus?

- A)** The protein is synthesised on ribosomes attached to the Golgi membrane itself and released directly into its cisternae.
- B)** The Golgi directs proteins only to lysosomes and the nucleus, while secreted proteins bypass it entirely.
- C)** Transport vesicles from the endoplasmic reticulum fuse with the cis face, the protein is modified and sorted as it moves through the cisternae, and secretory vesicles bud from the trans face.
- D)** Vesicles from the endoplasmic reticulum enter the Golgi at the trans face and leave from the cis face once modification is complete.
- E)** The Golgi returns all processed proteins to the endoplasmic reticulum, which then exports them from the cell.

Answer: C

Vesicles from the endoplasmic reticulum fuse with the cis face of the Golgi; the protein is chemically modified (chiefly its sugar chains) and sorted as it passes through the cisternae; and secretory vesicles bud from the trans face toward the plasma membrane. Secreted proteins are synthesised on ribosomes of the rough ER, not on the Golgi membrane, and the Golgi ships its products forward rather than returning everything to the ER for export.

23. In a gland cell that is actively exporting proteins, which statement about ribosomes is true?

- A)** Each ribosome is permanently bound or permanently free for its whole life, a status fixed when it is assembled in the nucleolus.
- B)** Bound ribosomes are structurally larger than free ribosomes, and only bound ribosomes can translate mRNA.
- C)** Secreted proteins are translated inside the Golgi apparatus, so ribosomes take no part in their synthesis.
- D)** Free ribosomes synthesise all proteins destined for export, which are then imported into the rough ER after they have folded.
- E)** Free ribosomes and ribosomes bound to the rough ER are structurally the same, and a ribosome binds to the ER because the protein it starts to translate carries a signal sequence.

Answer: E

Free ribosomes and ribosomes bound to the rough ER are structurally identical; a ribosome becomes bound only when the growing polypeptide's signal sequence is recognised at the ER membrane, so the same ribosome can be free in one round of translation and bound in the next, and its status is not fixed in the nucleolus. Secreted proteins are threaded into the ER while being translated, not imported after folding in the cytosol, and no translation occurs inside the Golgi.

24. Which of the following statements about the sodium potassium pump (Na^+/K^+ ATPase) of animal cells is true?

- A)** Its net effect is to make the inside of the cell electrically positive with respect to the outside.
- B)** It moves two Na^+ out of the cell and three K^+ in for each ATP hydrolysed, so it makes the interior slightly positive.
- C)** It carries Na^+ and K^+ down their concentration gradients and spends ATP only to make the carriers cycle faster.
- D)** It hydrolyses one ATP for each cycle in which it moves three Na^+ out of the cell and two K^+ in, maintaining gradients that secondary active transport and nerve impulses depend on.
- E)** It is a ligand gated channel that opens to let Na^+ and K^+ diffuse through whenever ATP binds to it.

Answer: D

For each ATP hydrolysed, the pump exports three Na^+ out of the cell and imports two K^+ , both against their concentration gradients, keeping Na^+ low and K^+ high inside and making the interior slightly more negative than the exterior. Those stored gradients power secondary active transport, such as glucose absorption in the intestine, and underlie the resting and action potentials of neurons. Moving ions down their gradients needs no ATP, an exchange of two out for three in would make the interior positive, which is the reverse of the facts, and the pump is an ATP-driven carrier, not a ligand-gated channel.

25. In blood passing through the capillaries of a vigorously contracting muscle, the local PCO_2 and the H^+ concentration both rise. What effect does this have on haemoglobin?

- A)** Haemoglobin's affinity for oxygen falls, the dissociation curve shifts to the right, and at any given PO_2 a larger fraction of the bound oxygen is unloaded to the muscle.
- B)** The heat produced by the muscle shifts the curve to the left and thereby cancels out the effect of the extra CO_2 .
- C)** Haemoglobin's affinity for oxygen rises, the dissociation curve shifts to the left, and oxygen is released only at very low PO_2 values.
- D)** Affinity is unchanged; the extra CO_2 simply displaces oxygen from the haeme iron in a one for one exchange.
- E)** The affinity falls because H^+ competes with oxygen for binding to the iron atoms of the haeme groups.

Answer: A

Rising PCO_2 and H^+ concentration lower haemoglobin's affinity for oxygen, the Bohr effect: the dissociation curve shifts to the right, so at any given PO_2 a larger fraction of the bound oxygen is unloaded to the working muscle, and the heat the muscle produces shifts the curve in the same (rightward) direction. H^+ binds to amino acid side chains and stabilises haemoglobin's low-affinity state; it does not compete with oxygen for the haeme iron, which remains the oxygen binding site, and a leftward shift with higher affinity is the opposite of what actually happens.

26. A patient spends four months completely immobile in bed. Which statement best describes what happens to bone remodelling in the unloaded bones?

- A)** Bone mass is controlled only by calcium regulating hormones, which are unaffected by loading, so no change occurs.
- B)** Deposition by osteoblasts rises to compensate for the missing load, so the bones become denser than before.
- C)** Resorption by osteoclasts exceeds deposition by osteoblasts, so bone mass falls, because mechanical loading is a normal stimulus for bone formation.
- D)** Osteoclasts are inactivated by disuse, while osteoblasts convert compact bone into spongy bone of equal mass.
- E)** Osteoblasts and osteoclasts both stop working, so bone mass remains exactly constant.

Answer: C

Mechanical loading is a normal stimulus for osteoblast deposition, so in unloaded bones resorption by osteoclasts exceeds deposition and bone mass falls, as seen after months of bed rest and in astronauts. Disuse does not silence osteoclasts, deposition cannot rise to compensate for a missing stimulus, and hormonal calcium regulation operates alongside mechanical regulation rather than replacing it, so bone mass does not stay constant.

27. Which of the following statements about the adrenal medulla is true?

- A)** It is regulated by releasing hormones from the hypothalamus, which stimulate the anterior pituitary to act on it.
- B)** Its hormones slow the heart and promote glycogen storage, preparing the body for rest and digestion.
- C)** It secretes steroid hormones that diffuse into target cells and bind to nuclear receptors.
- D)** Its main secretion, cortisol, raises blood glucose during prolonged stress.
- E)** It is directly innervated by preganglionic sympathetic neurons, so its hormones can be released into the blood within seconds of a stressful stimulus, without any pituitary hormone intervening.

Answer: E

The adrenal medulla behaves like a modified sympathetic ganglion: preganglionic sympathetic neurons innervate its chromaffin cells directly, so adrenaline and noradrenaline can be released into the blood within seconds of a stressful stimulus with no pituitary hormone intervening. Cortisol is a steroid secreted by the adrenal cortex under hypothalamus and anterior pituitary control, and medullary catecholamines act through cell-surface receptors to raise heart rate and blood glucose for fight or flight, not to promote rest and glycogen storage.

28. Which of the following is a normal function of the human liver?

- A)** Converting the ammonia released by amino acid breakdown into urea, which the blood carries to the kidneys for excretion.
- B)** Manufacturing red blood cells throughout adult life.
- C)** Producing insulin and glucagon in response to changing blood glucose.
- D)** Secreting digestive enzymes into the duodenum through the bile ducts.
- E)** Storing the bile produced by the gallbladder until fat reaches the small intestine.

Answer: A

The liver detoxifies the ammonia released by deamination of amino acids by converting it into urea, which the blood carries to the kidneys for excretion. Digestive enzymes are secreted by the pancreas, insulin and glucagon come from the pancreatic islets, red blood cells in adults are made in the bone marrow, and the liver produces bile while the gallbladder, not the liver, stores it.

29. Which of the following statements about a triglyceride molecule is true?

- A)** In a saturated triglyceride the fatty acid chains contain double bonds that introduce kinks and lower the melting point.
- B)** It consists of one glycerol joined by ester bonds to three fatty acids, and it is so non polar that it is essentially insoluble in water.
- C)** Its fatty acids are attached to glycerol by peptide bonds, so dietary fat is digested by proteases.
- D)** It is built from three glycerol molecules condensed onto a single fatty acid chain.
- E)** It carries a phosphate group in place of one fatty acid, which is why triglycerides form the bilayers of cell membranes.

Answer: B

A triglyceride is one glycerol molecule joined by ester bonds to three fatty acids, and with no charged or strongly polar head group it is non-polar and essentially insoluble in water. Replacing one fatty acid with a phosphate group gives a phospholipid, the membrane-forming lipid, not a triglyceride; ester bonds are digested by lipases, not proteases; and saturated fats by definition have no double bonds, so no kinks or lowered melting point. The molecule is one glycerol with three acids, not three glycerols on one acid.

30. Which statement correctly describes quiet exhalation at rest?

- A)** Air leaves the lungs because alveolar pressure falls below atmospheric pressure during exhalation.
- B)** The diaphragm flattens further during exhalation, pressing the abdominal organs upward against the lungs.
- C)** Quiet exhalation is powered by contraction of the internal intercostal muscles and the abdominal muscles, which squeeze the air out of the lungs.
- D)** The diaphragm and external intercostal muscles relax, the lungs and chest wall recoil elastically, alveolar pressure rises above atmospheric pressure, and air flows out with no active muscle contraction.
- E)** Quiet exhalation uses exactly the same muscles as quiet inhalation, but contracts them in the reverse direction.

Answer: D

Quiet exhalation is passive: the diaphragm and external intercostal muscles relax, the elastic recoil of the lungs and chest wall reduces lung volume, alveolar pressure rises above atmospheric pressure, and air flows out with no active contraction. The internal intercostals and abdominal muscles are recruited only for forced exhalation; air always flows from higher to lower pressure, so a fall of alveolar pressure below atmospheric would draw air in; and muscles cannot contract in a reverse direction.

31. In a plant species, tall stem (T) is dominant over dwarf (t). A tall plant of unknown genotype is crossed with a dwarf plant, and the offspring consist of 61 tall and 59 dwarf plants. What conclusion follows?

- A)** The tall parent is heterozygous (Tt), because crossing a heterozygote with a homozygous recessive plant yields a 1:1 phenotype ratio, reflecting segregation of T and t into equal numbers of gametes.
- B)** The tall parent is homozygous TT, and the dwarf offspring must have arisen by new mutation.
- C)** The dwarf parent must actually be heterozygous, since half of the offspring grew tall.
- D)** The 1:1 ratio indicates incomplete dominance, the heterozygotes being of intermediate height.
- E)** The genotype of the tall parent cannot be deduced from the phenotypes of the offspring.

Answer: A

The dwarf parent must be tt and contributes only t gametes, so an offspring is tall only if it receives T from the tall parent. The 61 tall to 59 dwarf offspring, about 1:1, therefore mean the tall parent makes T and t gametes in equal numbers, that is, it is heterozygous Tt, the classic test cross. A TT parent would give all tall offspring, the dwarf parent cannot be heterozygous because dwarf is the recessive phenotype, and a 1:1 phenotype ratio reflects equal segregation of gametes, not incomplete dominance.

32. Most mature neurons of an adult human never divide again, whereas liver cells can resume division after part of the liver is surgically removed. Which statement explains this difference?

- A)** Liver cells regenerate the organ by meiosis, followed by fusion of the cells formed.
- B)** Neurons remain permanently arrested at the metaphase checkpoint, with chromosomes aligned but unable to separate.
- C)** Both cell types are outside the cycle, and liver regrowth occurs purely by enlargement of existing cells, with no new cell divisions.
- D)** Neurons keep replicating their DNA but cannot condense their chromosomes, so they become polyploid.
- E)** Neurons are held in a permanent G0 phase outside the division cycle, while liver cells remain able to re-enter the cycle from G0 when growth signals arrive.

Answer: E

Mature neurons have permanently exited the cell cycle into the G0 phase and never divide again, whereas hepatocytes remain able to re-enter G1 and proliferate when growth signals arrive after partial hepatectomy, which is how the liver regrows. Neurons are not arrested at a metaphase checkpoint and do not keep replicating DNA into polyploidy, and liver regeneration occurs by mitotic division of existing cells, not by meiosis, cell fusion, or mere enlargement of cells without division.

Section C - Chemistry (Q33-47)

33. Acidified potassium dichromate(VI) solution oxidises iodide ions to iodine: $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{I}^-(\text{aq}) \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{I}_2(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$. Which of the following statements is correct?

- A)** Chromium is oxidised from +6 to +7, while iodine is reduced from 0 to -1.
- B)** Iodine is oxidised, its oxidation number rising from -1 to 0, while chromium is reduced from +6 to +3.
- C)** Neither element changes its oxidation number, because this is not a redox reaction.
- D)** Iodine is reduced from 0 to -1, while chromium is oxidised from +3 to +6.
- E)** Iodine is oxidised, its oxidation number rising from -1 to 0, while chromium is reduced from +7 to +3.

Answer: B

In the dichromate ion, $2x + 7(-2) = -2$ gives $x = +6$ for each chromium; in Cr^{3+} chromium is +3, so chromium is reduced from +6 to +3. Iodide goes from -1 in I^- to 0 in I_2 , an increase, so iodine is the element oxidised. The +7 figure comes from ignoring the 2- charge of the dichromate ion, and the reaction is genuinely redox because both elements change oxidation number.

34. A coil of copper wire is immersed in a solution of silver nitrate, and the reaction $\text{Cu}(\text{s}) + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$ slowly produces a blue solution and silvery crystals on the wire. Which of the following statements is correct?

- A)** Cu is the oxidising agent because it donates two electrons.
- B)** Cu^{2+} is the oxidising agent because it carries the larger charge.
- C)** Ag is the reducing agent because it accepts electrons from the copper.
- D)** Ag^+ is the oxidising agent because it accepts electrons from copper atoms.
- E)** NO_3^- is reduced to NO while the silver is deposited.

Answer: D

Each copper atom loses two electrons (oxidation number 0 to +2), so copper is oxidised and acts as the reducing agent; each Ag^+ ion gains one of those electrons (+1 to 0), so Ag^+ is reduced and is therefore the oxidising agent, which by definition is the species that accepts electrons. Nitrate ions are spectators and are unchanged, the size of an ion's charge does not make it an oxidising agent, and the silver metal produced has already been reduced, so it cannot be the reducing agent in this reaction.

35. According to VSEPR theory, what are the shape of the ammonia molecule, NH_3 , and the approximate H-N-H bond angle?

- A)** tetrahedral, about 109.5 degrees
- B)** bent, about 104.5 degrees
- C)** trigonal pyramidal, about 107 degrees
- D)** linear, about 180 degrees
- E)** trigonal planar, about 120 degrees

Answer: C

Nitrogen in NH_3 has four electron domains, three bonding pairs and one lone pair, arranged roughly tetrahedrally. The lone pair repels more strongly than the bonding pairs, compressing the H-N-H angle below 109.5 degrees, so the molecule is a trigonal pyramid with an angle of about 107 degrees. The 104.5 degrees value belongs to water, which has two lone pairs, while 120 or 180 degrees would require trigonal planar or linear electron arrangements, which NH_3 does not have.

36. Water (Mr 18) boils at 100 °C, whereas hydrogen sulfide (Mr 34) boils at about -60 °C. Which statement explains this difference?

- A)** Water molecules are held together by ionic bonds between H⁺ and OH⁻ ions.
- B)** Water molecules attract one another through hydrogen bonds, which are much stronger than the dipole-dipole and London forces present in liquid hydrogen sulfide.
- C)** Water boils higher because breaking the strong O-H covalent bonds inside its molecules requires more energy than breaking S-H bonds.
- D)** Hydrogen sulfide molecules form hydrogen bonds with each other, while water molecules cannot.
- E)** Hydrogen sulfide in fact boils higher than water because it has the larger molar mass.

Answer: B

Boiling separates molecules, so only intermolecular forces matter. Water's hydrogen atoms are bonded to the very electronegative oxygen, allowing strong hydrogen bonds between molecules, whereas sulfur in H₂S is not electronegative enough for hydrogen bonding, so H₂S relies on much weaker dipole-dipole and London forces despite its larger molar mass. The covalent O-H bonds inside the molecule are not broken at boiling, water is not an ionic compound, and H₂S in fact boils about 160 degrees lower than water, not higher.

37. 10 cm³ of a 1.0×10^{-2} mol/dm³ solution of hydrochloric acid, which is fully ionised in water, is diluted with water to 100 cm³. Then 10 cm³ of this diluted solution is further diluted with water to 1000 cm³. What is the pH of the final solution?

- A)** 2 **B)** 3 **C)** 4 **D)** 5 **E)** 9

Answer: D

Step 1: the first dilution takes 10 cm³ to 100 cm³, a tenfold dilution, so the concentration becomes $1.0 \times 10^{-2} / 10 = 1.0 \times 10^{-3}$ mol/dm³. Step 2: the second dilution takes 10 cm³ to 1000 cm³, a hundredfold dilution, so the concentration becomes $1.0 \times 10^{-3} / 100 = 1.0 \times 10^{-5}$ mol/dm³. Step 3: HCl is fully ionised, so $[H^+] = 1.0 \times 10^{-5}$ mol/dm³ and $pH = -\log(1.0 \times 10^{-5}) = 5$. A pH of 3 accounts for only the first dilution, and 9 is the pOH (14 - 5), not the pH.

38. Aqueous ethanoic acid is at equilibrium: $CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$. The solution is diluted with pure water at constant temperature. Which of the following statements is correct?

- A)** The value of K_a decreases in proportion to the dilution.
- B)** The degree of dissociation decreases, because there are fewer acid molecules per unit volume.
- C)** The pH falls, because dilution causes more H⁺ ions to be produced.
- D)** The equilibrium shifts to the left, towards undissociated CH₃COOH.
- E)** The degree of dissociation increases, but the concentrations of all species fall, so the pH rises.

Answer: E

Dilution lowers the concentrations of all dissolved species, so by Le Chatelier's principle the equilibrium shifts toward the side with more dissolved particles, the ion side, meaning the degree of dissociation increases. The extra dissociation cannot fully offset the dilution, so $[H^+]$ still falls and the pH rises. K_a depends only on temperature and is unchanged, a decrease in the degree of dissociation or a fall of pH is the opposite of what happens, and the shift is toward the ions, not toward undissociated CH₃COOH.

39. Serine is an alpha-amino acid whose side chain is $\text{-CH}_2\text{OH}$. Which set of functional groups is present in a molecule of serine?

- A)** amino, carboxyl and hydroxyl
- B)** carboxyl, hydroxyl and ester
- C)** amino, hydroxyl and keto
- D)** amino, carboxyl and amide
- E)** amino, carboxyl and thiol

Answer: A

The backbone of every alpha-amino acid carries an amino group and a carboxyl group attached to the alpha carbon, and serine's side chain, $\text{-CH}_2\text{OH}$, is an alcohol, contributing a hydroxyl group. A thiol side chain would make the amino acid cysteine and an amide side chain asparagine, while no ester or keto group appears anywhere in serine's structure.

40. But-1-ene and but-2-ene are structural isomers with the molecular formula C_4H_8 . Which of the following statements is correct?

- A)** Neither alkene shows cis-trans isomerism because their $\text{C}=\text{C}$ bonds rotate freely.
- B)** Both alkenes can exist as cis and trans isomers.
- C)** Only but-2-ene can exist as cis and trans isomers, because in but-1-ene one carbon of the $\text{C}=\text{C}$ double bond carries two identical hydrogen atoms.
- D)** But-2-ene shows cis-trans isomerism because rotation about its $\text{C}=\text{C}$ double bond is completely free.
- E)** Only but-1-ene shows cis-trans isomerism, because its double bond is at the end of the carbon chain.

Answer: C

Cis-trans isomerism about a $\text{C}=\text{C}$ bond requires each double-bond carbon to carry two different groups, since rotation about the double bond is restricted. In but-1-ene, $\text{CH}_2=\text{CH-CH}_2\text{-CH}_3$, the terminal double-bond carbon carries two identical hydrogen atoms, so no distinct cis and trans forms exist; in but-2-ene each double-bond carbon carries one H and one CH_3 , so both cis and trans forms exist. Claiming free rotation about a $\text{C}=\text{C}$ bond contradicts the very existence of cis-trans isomers, and it is but-2-ene, not but-1-ene, that shows them.

41. Sodium, magnesium and aluminium are consecutive elements in the third period of the periodic table. Which statement is correct?

- A)** Their atomic radii increase from sodium to aluminium because each atom contains more electrons.
- B)** Sodium has the smallest atomic radius of the three because it has the fewest electrons.
- C)** Their atomic radii decrease from sodium to aluminium because nuclear charge increases while the outer electrons remain in the same shell.
- D)** Their first ionisation energies decrease from sodium to aluminium because the atoms become heavier.
- E)** Aluminium is the least electronegative of the three because metallic character increases across a period.

Answer: C

Across period 3 the nuclear charge rises from +11 for sodium to +12 for magnesium to +13 for aluminium while the outer electrons remain in the same third shell, so the stronger pull draws the shell inward and the atomic radius decreases from sodium to aluminium. First ionisation energy and electronegativity both increase across the period, so aluminium is the most, not the least, electronegative of the three and metallic character falls, and it is sodium that has the largest radius, not the smallest.

42. A fixed mass of an ideal gas occupies 6.0 dm³ at 27 °C and 150 kPa. What volume will the same mass of gas occupy at 127 °C and 100 kPa?

- A)** 3.0 dm³ **B)** 6.0 dm³ **C)** 8.0 dm³ **D)** 9.0 dm³ **E)** 12.0 dm³

Answer: E

Step 1: convert temperatures to kelvin: $T_1 = 27 + 273 = 300 \text{ K}$ and $T_2 = 127 + 273 = 400 \text{ K}$

Step 2: apply the combined gas law, $V_2 = V_1 \times (T_2/T_1) \times (P_1/P_2) = 6.0 \times (400/300) \times (150/100) = 6.0 \times 4/3 \times 3/2 = 12.0 \text{ dm}^3$. Heating the gas at fixed pressure would raise the volume to 8.0 dm³ and halving the pressure at fixed temperature would raise it to 9.0 dm³, so only applying both changes gives 12.0 dm³, while inverting either ratio gives a wrongly smaller volume.

43. Nitrogen dioxide dimerises: $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$, with $\Delta H = -57 \text{ kJ/mol}$ for the forward reaction. An equilibrium mixture of the two gases is heated at constant volume. Which statement is correct?

- A)** The equilibrium shifts towards NO₂ and the value of K_c decreases.
B) The equilibrium shifts towards N₂O₄ and the value of K_c stays the same.
C) The equilibrium shifts towards NO₂ and the value of K_c increases.
D) The equilibrium shifts towards N₂O₄ and the value of K_c increases.
E) The position of equilibrium does not change; only the rates of the forward and reverse reactions increase.

Answer: A

The forward reaction (dimerisation to N₂O₄) is exothermic, so raising the temperature favours the endothermic reverse direction and the equilibrium shifts toward NO₂. K_c changes only with temperature, and for an exothermic forward reaction its value decreases on heating. A shift toward N₂O₄ contradicts Le Chatelier's principle, and heating at constant volume is precisely a temperature change, so it cannot leave the position of equilibrium or K_c unchanged.

44. Which of the following statements about the bonding in benzene, C₆H₆, is correct?

- A)** All six C-C bonds in benzene are single bonds, and the sixth valence electron of each carbon moves freely like the electrons of a metal.
B) The six carbon atoms form a planar ring with six delocalised pi electrons spread over the whole ring, making all six C-C bonds equivalent and intermediate in length between single and double bonds.
C) The benzene ring is non-planar, with its pi electrons localised on alternating carbon atoms.
D) Benzene undergoes addition in preference to substitution because addition preserves the delocalised system.
E) Benzene contains three separate C=C double bonds alternating with single bonds, which explains why it reacts mainly by addition.

Answer: B

Each of the six sp²-hybridised carbons contributes one electron to a pi system delocalised around the planar ring, so all six C-C bonds are equivalent and intermediate in length between single and double bonds (about 139 pm). Because the delocalised system stabilises the ring, benzene reacts mainly by substitution, which preserves it, rather than by addition, which would destroy it. A Kekule structure of three fixed alternating double bonds does not describe real benzene, and the ring is planar, not non-planar.

45. For a given reversible reaction, the value of the equilibrium constant K_c is changed by which of the following?

- A)** Increasing the total pressure.
- B)** Increasing the concentration of a reactant.
- C)** Adding a catalyst.
- D)** A change in temperature.
- E)** Adding an inert gas at constant volume.

Answer: D

The value of K_c depends only on temperature, so only a temperature change alters it. Changing reactant concentrations or total pressure, or adding an inert gas at constant volume, can shift the position of equilibrium but leaves K_c unchanged, and a catalyst accelerates the forward and reverse reactions equally, changing neither the position of equilibrium nor K_c .

46. What is the chemical formula of iron(III) sulfate?

- A)** FeSO_4 **B)** $\text{Fe}(\text{SO}_4)_3$ **C)** $\text{Fe}_2(\text{SO}_4)_3$ **D)** $\text{Fe}_3(\text{SO}_4)_2$ **E)** Fe_2S_3

Answer: C

Iron(III) is the Fe^{3+} ion and sulfate is the SO_4^{2-} ion; the least common multiple of the charges is 6, so two Fe^{3+} ions balance three sulfate ions, giving $\text{Fe}_2(\text{SO}_4)_3$. FeSO_4 is iron(II) sulfate, $\text{Fe}(\text{SO}_4)_3$ and $\text{Fe}_3(\text{SO}_4)_2$ leave the charges unbalanced, and Fe_2S_3 wrongly treats sulfate as if it were bare sulfide.

47. Naturally occurring potassium contains the radioactive isotope potassium-40. How many neutrons are there in one atom of potassium-40? (Potassium has atomic number 19.)

- A)** 19 **B)** 20 **C)** 21 **D)** 40 **E)** 59

Answer: C

The number of neutrons equals the mass number minus the atomic number: $40 - 19 = 21$. The value 19 is the number of protons, 40 is the mass number itself, and 20 neutrons would correspond to potassium-39, the common stable isotope, not to radioactive potassium-40.

Section D - Mathematics (Q48-54)

48. Solve the inequality $\sqrt{2x - 5} < 9$.

- A)** $x \leq 43$ **B)** $x < 43$ **C)** $x \geq 5/2$ **D)** $5/2 \leq x < 43$ **E)** $5/2 < x < 43$

Answer: D

Step 1: the square root requires $2x - 5 \geq 0$, so $x \geq 5/2$

Step 2: both sides are non-negative on the domain, so squaring preserves the inequality: $2x - 5 < 81$, giving $x < 43$.

Step 3: combining, $5/2 \leq x < 43$

Omitting the domain condition gives $x < 43$, which wrongly admits values below $5/2$, and the endpoint $x = 5/2$ is included because $\sqrt{0} = 0 < 9$.

49. Solve the inequality $\cos(x) \leq 1/2$ for $0 \leq x \leq 2\pi$.

- A) $2\pi/3 \leq x \leq 4\pi/3$
- B) $\pi/3 \leq x \leq 5\pi/3$
- C) $0 \leq x \leq \pi/3$ or $5\pi/3 \leq x \leq 2\pi$
- D) $\pi/3 \leq x \leq 2\pi/3$
- E) $\pi/3 < x < 5\pi/3$

Answer: B

$\cos(x) = 1/2$ at $x = \pi/3$ and $x = 5\pi/3$, and since cosine decreases from 1 to -1 on $[0, \pi]$ and rises again from π to 2π , the values with $\cos(x) \leq 1/2$ lie between these two crossings, with $\cos(\pi) = -1$ confirming the middle of the interval qualifies.

The solution set is therefore $\pi/3 \leq x \leq 5\pi/3$

Removing the endpoints excludes the values where equality holds, and the two outer intervals describe exactly the region where $\cos(x) > 1/2$, the opposite of what is asked.

50. In a survey of 100 tourists visiting Rome, 62 say they will visit the Colosseum, 55 say they will visit the Vatican Museums, and 17 say they will visit neither of the two. How many of the tourists will visit both attractions?

- A) 17 B) 28 C) 34 D) 38 E) 117

Answer: C

Step 1: 17 tourists visit neither attraction, so $100 - 17 = 83$ visit at least one of the two. Step 2: by inclusion-exclusion, the number visiting at least one equals $62 + 55$ minus the number visiting both, so the overlap is $62 + 55 - 83 = 34$. Adding the two counts without subtracting the overlap gives 117, which exceeds the total of 100 tourists, and omitting the neither group from the calculation gives the wrong overlap.

51. The polynomial $P(x) = x^3 + ax + b$ is exactly divisible by $(x - 1)$, and when $P(x)$ is divided by $(x + 2)$ the remainder is 6. What is the value of a ?

- A) -5 B) -3 C) 1 D) 4 E) 5

Answer: A

Step 1: divisibility by $(x - 1)$ means $P(1) = 0$ by the factor theorem: $1 + a + b = 0$. Step 2: the remainder on division by $(x + 2)$ is $P(-2) = 6$ by the remainder theorem: $-8 - 2a + b = 6$. Step 3: subtracting the second equation from the first gives $9 + 3a = -6$, so $a = -5$ and then $b = 4$.

Check: $P(x) = x^3 - 5x + 4$ gives $P(1) = 0$ and $P(-2) = -8 + 10 + 4 = 6$, as required

52. A triangle has perimeter 72 cm and its side lengths are in the ratio 3 : 4 : 5. What is the area of the triangle?

- A) 108 cm² B) 216 cm² C) 270 cm² D) 360 cm² E) 432 cm²

Answer: B

Step 1: the perimeter of 72 cm is shared among $3 + 4 + 5 = 12$ ratio parts, each part $72/12 = 6$ cm, giving sides 18, 24 and 30 cm. Step 2: since $18^2 + 24^2 = 324 + 576 = 900 = 30^2$, the triangle is right angled with legs 18 and 24.

Step 3: area = $(18 \times 24)/2 = 216$ cm²

Using the hypotenuse 30 as the base with a leg as height, or forgetting to halve the product of the legs, produces the larger distractor values.

53. Marco is 26 years old and his sister Elisa is 14 years old. After how many years will Marco be exactly one and a half times as old as Elisa?

- A) 2 B) 5 C) 10 D) 12 E) 24**

Answer: C

Let x be the number of years.

Then $26 + x = 1.5(14 + x)$, so $26 + x = 21 + 1.5x$, which gives $0.5x = 5$ and $x = 10$

Check: after 10 years Marco is 36 and Elisa is 24, and $36 = 1.5 \times 24$.

54. Solve the equation $3^{(x+3)} = 9^{(x-1)}$.

- A) -5 B) 1 C) 3 D) 5 E) 7**

Answer: D

Write both sides as powers of 3: $9^{(x-1)} = (3^2)^{(x-1)} = 3^{(2x-2)}$

Equating the exponents of the equal powers of 3 gives $x + 3 = 2x - 2$, so $x = 5$.

Check: $3^{(5+3)} = 3^8 = 6561$ and $9^{(5-1)} = 9^4 = 6561$

Section E - Physics (Q55-60)

55. On a straight motorway a car travels at a constant 108 km/h. A truck ahead travels in the same direction at a constant 25 m/s. At one instant the truck is 100 m in front of the car. How long does it take the car to draw level with the truck?

- A) 3.3 s B) 4 s C) 10 s D) 20 s E) 33 s**

Answer: D

Step 1: convert the car's speed: $108 \text{ km/h} = 108/3.6 = 30 \text{ m/s}$

Step 2: since both vehicles move in the same direction, the gap closes at the relative speed $30 - 25 = 5 \text{ m/s}$.

Step 3: the 100 m gap closes in $t = 100/5 = 20 \text{ s}$

Using the car's full speed as the closing speed, ignoring the truck, would give about 3.3 s, which is far too short because the truck is also moving forward.

56. A 60 V battery of negligible internal resistance is connected to an 8Ω resistor in series with a parallel combination of a 20Ω resistor and a 30Ω resistor. What current does the battery supply?

- A) 2 A B) 3 A C) 4 A D) 5 A E) 7.5 A**

Answer: B

Step 1: the parallel combination of 20Ω and 30Ω has equivalent resistance $(20 \times 30)/(20 + 30) = 600/50 = 12 \Omega$.

Step 2: the total circuit resistance is $8 + 12 = 20 \Omega$

Step 3: $I = V/R = 60/20 = 3 \text{ A}$

Adding 20Ω and 30Ω as if they were in series would give 50Ω and a much smaller current, and forgetting the 8Ω series resistor would give $60/12 = 5 \text{ A}$.

57. A current is passed through molten potassium chloride held between two electrodes. Which one of the following statements is correct?

- A)** No current can flow, because the ions of a molten salt are not free to move.
- B)** The positive ions and the negative ions drift in the same direction.
- C)** The K^+ ions drift toward the positive electrode and the Cl^- ions toward the negative electrode.
- D)** The conventional current in the liquid is carried only by the negative ions.
- E)** The direction of the conventional current in the liquid is the direction of drift of the positive ions.

Answer: E

In the molten salt, K^+ ions drift toward the negative electrode and Cl^- ions toward the positive electrode, with both species contributing to the current. Conventional current is defined as the direction of flow of positive charge, so inside the liquid it points along the drift of the K^+ ions, which is the correct statement. Swapping the destinations of the two ions reverses the physics, the two ion species drift in opposite directions rather than the same one, and molten salts conduct precisely because their ions are free to move.

58. The acceleration due to gravity at the surface of a planet of radius R is g . At what height h above the surface has the gravitational acceleration fallen to exactly one quarter of its surface value?

- A)** R **B)** $3R$ **C)** $R/4$ **D)** $2R$ **E)** $R/2$

Answer: A

Gravitational acceleration follows $g = GM/r^2$, with r measured from the planet's centre, so g falls as the inverse square of the distance. For g to fall to one quarter of its surface value, the distance from the centre must double: $r = 2R$.

The height above the surface is therefore $h = 2R - R = R$

Answering $2R$ forgets that r already includes the planet's radius, and $R/4$ or $R/2$ treat the falloff as linear rather than inverse-square.

59. A 10 kg box is pulled along a horizontal floor by a horizontal force of 60 N. The coefficient of kinetic friction between the box and the floor is 0.40. What is the acceleration of the box? ($g = 10 \text{ m/s}^2$)

- A)** 0.4 m/s^2 **B)** 2 m/s^2 **C)** 5.6 m/s^2 **D)** 6 m/s^2 **E)** 20 m/s^2

Answer: B

Step 1: the normal force is $N = mg = 10 \times 10 = 100 \text{ N}$, so kinetic friction is $f = 0.40 \times 100 = 40 \text{ N}$, opposing the motion.

Step 2: the net force is $60 - 40 = 20 \text{ N}$

Step 3: $a = F/m = 20/10 = 2 \text{ m/s}^2$

Ignoring friction would give $60/10 = 6 \text{ m/s}^2$, mistaking the friction coefficient for an acceleration gives 0.4 m/s^2 , and quoting the net force of 20 N as if it were an acceleration gives 20 m/s^2 .

60. A rectangular drain hatch of area 0.5 m^2 lies flat on the bottom of a swimming pool filled to a depth of 4.0 m with water of density 1000 kg/m^3 . What force does the water exert on the hatch? ($g = 10 \text{ m/s}^2$; ignore atmospheric pressure.)

- A)** 2000 N **B)** 4000 N **C)** 20000 N **D)** 40000 N **E)** 80000 N

Answer: C

Step 1: the gauge pressure at depth h in a liquid is $P = \rho \times g \times h = 1000 \times 10 \times 4.0 = 40000 \text{ Pa}$. Step 2: the force on the hatch is pressure times area $= 40000 \times 0.5 = 20000 \text{ N}$. Quoting 40000 confuses the pressure in pascals with the force in newtons, and 80000 would follow from failing to multiply by the area; atmospheric pressure is explicitly to be ignored, and including it would give a much larger value.